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FORM 5

SEMESTRAL DISTRIBUTION OF COURSES

A. Non-Scholar

MASTER OF SCIENCE IN COMPUTER APPLICATIONS

(LIST OF COURSES BY SEMESTER)

BOR Resolution No. 128 , Series of (2023)

Course Number	Course Title	Units	Hours/Week			Pre-req	
			Lec	Lab	Total		
First Year - First Semester							
	MCA201	Advanced Computer Organization	3	3	0	3	
	MCA202	Advanced Operating Systems	3	3	0	3	
	CCC297	Methods of Research	3	3	0	3	
		Elective Course 1	3	3	0	3	
		TOTAL	12			12	
First Year - Second Semester							
		Elective Course 2	3	3	0	3	
		Elective Course 3	3	3	0	3	
	MCA203	System Development with Emerging Technologies	3	3	0	3	
	CCC296	ICT for Peace and Development	1	1	0	1	
		TOTAL	10			10	
Second Year - First Semester							
	CCC298	Systematic Review and Research Seminar in ICT	3	3	0	3	
		TOTAL	3			3	
Second Year - Second Semester							
	MCA300	Master’s Thesis	6	6	0	6	
		TOTAL	6			6	
		GRAND TOTAL	31			31	
Note: Comprehensive Examination will be taken during the first-year summer term after gaining 12 units with the required core subjects and enroll in Graduate Residency (GRR) with 0 units if they are not enrolled in any other course.							



B. ERDT Scholarship

MASTER OF SCIENCE IN COMPUTER APPLICATIONS
(LIST OF COURSES BY SEMESTER)

BOR Resolution No. 128 , Series of (2023)

Course Number	Course Title	Units	Hours/Week			Pre-req	
			Lec	Lab	Total		
First Year - First Semester							
MCA201	Advanced Computer Organization	3	3	0	3		
MCA202	Advanced Operating Systems	3	3	0	3		
CCC297	Methods of Research	3	3	0	3		
	Elective Course 1	3	3	0	3		
TOTAL		12			12		
First Year - Second Semester							
	Elective Course 2	3	3	0	3		
	Elective Course 3	3	3	0	3		
MCA203	System Development with Emerging Technologies	3	3	0	3		
CCC296	ICT for Peace and Development	1	1	0	1		
TOTAL		10			10		
First Year - Summer							
ENS 293	Technology Entrepreneurship	3	3	0	3		
TOTAL		3			3		
Second Year - First Semester							
CCC298	Systematic Review and Research Seminar in ICT	3	3	0	3		
TOTAL		3			3		
Second Year - Second Semester							
MCA300	Master’s Thesis	6	6	0	6		
TOTAL		6			6		
GRAND TOTAL		34			34		
Note: Comprehensive Examination will be taken during the first-year summer term after gaining 12 units with the required core subjects and enroll in Graduate Residency (GRR) with 0 units if they are not enrolled in any other course.							



C. Graduate Student with Bridging Course

MASTER OF SCIENCE IN COMPUTER APPLICATIONS
(LIST OF COURSES BY SEMESTER)

BOR Resolution No. 128 , Series of (2023)

Course Number	Course Title	Units	Hours/Week			Pre-req
			Lec	Lab	Total	
First Year - First Semester						
MCA201	Advanced Computer Organization	3	3	0	3	
MCA202	Advanced Operating Systems	3	3	0	3	
CCC297	Methods of Research	3	3	0	3	
	Specialization Course 1	3	3	0	3	
Elective 1	Technical Elective 1	3	3	0	3	
	TOTAL	15			15	
First Year - Second Semester						
	Elective Course 2	3	3	0	3	
	Elective Course 3	3	3	0	3	
MCA203	System Development with Emerging Technologies	3	3	0	3	
CCC296	ICT for Peace and Development	1	1	0	1	
Elective 2	Technical Elective 2	3	3	0	3	
Elective 3	Technical Elective 3	3	3	0	3	
	TOTAL	16			16	
Second Year - First Semester						
CCC298	Systematic Review and Research Seminar in ICT	3	3	0	3	
Elective 4	Technical Elective 4	3	3	0	3	
	TOTAL	6			6	
Second Year - Second Semester						
MCA300	Master’s Thesis	6	6	0	6	
	TOTAL	6			6	
	GRAND TOTAL	43			43	
Note: Comprehensive Examination will be taken during the first-year summer term after gaining 12 units with the required core subjects and enroll in Graduate Residency (GRR) with 0 units if they are not enrolled in any other course.						



The following graduation requirements shall be observed for any candidate for graduation.

At least one (1) published article in a refereed journal or juried creative work outlet.

Specialization Courses	Course Title	Units
MCA221	Advanced Embedded Systems	3
MCA222	Advanced Digital Signal Processing	3
MCA223	Advanced Computer Vision	3
MCA224	Applied Machine Learning for Embedded Systems	3
MCA225	Advanced Internet of Things	3
MCA226	Network Security for IOT	3
MCA227	Advanced Cloud Computing	3
MCA228	Applied Big-Data & Analytics for IoT	3

Students with bridging courses will be enrolled from the following electives courses with a total maximum of 12 units, subject to the adviser program evaluation of study.

Elective Courses	Course Title	Units
CCC101	Computer Programming 1	3
BCA121	Fundamentals of Computer Applications 2	3
BCA180	Introduction of Computer Vision	3
BCA181	HDL Programming	3
BCA182	Embedded Systems Programming	3
BCA183	Digital Signal Processing	3
BCA188	Programming For Internet of Things	3
BCA189	Applied Internet of Things	3
BCA190	Artificial Intelligence in Internet of Things	3
BCA191	Fundamentals of Robotics Systems	3

The MSCA student may also enroll to other MS courses of other department or college when needed and with the consent of the adviser.